

DUNEDIN, New Zealand

WMO: 938910

Lat: 45.9267S

Lon: 170.1967E

Elev: 2

StdP: 101.30

Time Zone: 12.00 (NZL)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -4.3	(c) -3.0	(d) -4.8	(e) 2.5	(f) -3.0	(g) -3.6	(h) 2.8	(i) -1.9	(j) 12.1	(k) 9.1	(l) 10.7	(m) 8.8	(n) 0.7	(o) 200	(p) 0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 10.5	(c) 26.6	(d) 17.5	(e) 24.6	(f) 16.8	(g) 22.8	(h) 16.1	(i) 18.7	(j) 24.5	(k) 17.7	(l) 23.0	(m) 16.8	(n) 21.6	(o) 5.0	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.5	11.7	20.9	15.5	11.0	19.7	14.6	10.4	18.8	52.9	24.9	49.9	23.0	47.2	21.6	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.3	(b) 9.8	(c) 8.5	(e) -6.0	(f) 30.6	(g) 1.2	(h) 1.6	(i) -6.9	(j) 31.8	(k) -7.6	(l) 32.8	(m) -8.2	(n) 33.7	(o) -9.1	(p) 34.9		
(a) 11.3	(b) 9.8	(c) 8.5	(e) -6.2	(f) 20.8	(g) 1.2	(h) 1.0	(i) -7.1	(j) 21.5	(k) -7.9	(l) 22.1	(m) -8.6	(n) 22.6	(o) -9.5	(p) 23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	15.2	14.7	13.4	10.7	8.1	5.6	5.2	6.6	8.6	10.0	12.2	14.1		
	DBStd	4.48	3.00	2.98	3.10	2.89	2.84	2.75	2.82	2.56	2.77	2.77	3.09	3.07		
	HDD10.0	603	1	1	4	25	72	134	149	109	59	33	12	3		
	HDD18.3	2938	106	107	157	229	317	382	406	364	292	258	186	135		
	CDD10.0	731	162	133	109	46	14	2	2	4	17	34	77	131		
	CDD18.3	24	9	6	3	0	0	0	0	0	0	1	1	4		
	CDH23.3	316	122	72	37	2	0	0	0	0	0	6	21	57		
	CDH26.7	58	28	13	6	0	0	0	0	0	0	1	2	8		
(14) Wind	WSAvg	3.3	3.6	3.3	3.0	3.0	2.9	3.0	2.8	2.8	3.6	3.8	3.9	3.7		
(15) Precipitation	PrecAvg	698	70	57	56	55	70	55	50	50	46	62	58	68		
	PrecMax	940	185	109	159	159	234	161	130	129	105	136	121	110		
	PrecMin	528	21	17	12	10	13	12	15	16	16	12	17	17		
	PrecStd	102	41	28	29	31	50	39	29	25	19	28	26	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	28.7	27.0	22.8	19.8	16.6	15.7	17.4	21.3	23.7	26.1	27.8		
		MCWB	18.9	18.4	17.5	14.5	13.0	10.7	9.7	10.6	12.7	14.5	16.9	17.7		
	2%	DB	26.5	25.6	23.8	20.2	16.9	14.0	13.6	15.0	18.4	20.4	23.0	24.9		
		MCWB	17.4	17.6	16.5	13.5	11.3	9.6	8.8	9.8	11.6	13.2	15.2	17.2		
	5%	DB	24.1	23.0	21.6	18.2	15.1	12.1	11.9	13.3	16.4	18.1	20.9	22.5		
		MCWB	16.8	16.7	15.8	13.1	11.0	8.8	8.1	9.2	10.7	12.0	14.4	16.0		
	10%	DB	21.7	21.1	19.7	16.4	13.5	10.5	10.3	11.7	14.7	16.2	18.7	20.4		
		MCWB	15.9	16.0	14.8	12.3	10.1	8.0	7.4	8.5	10.0	11.3	13.4	15.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.1	20.3	19.0	15.7	14.0	11.5	10.5	11.3	13.7	15.7	17.9	19.0		
		MCDB	26.1	25.3	24.7	20.6	18.7	15.1	13.9	15.7	19.3	21.6	24.2	25.2		
	2%	WB	18.6	18.4	17.4	14.5	12.3	10.1	9.5	10.3	12.3	14.1	16.2	17.7		
		MCDB	24.6	23.5	22.2	18.4	15.6	12.9	12.5	13.9	17.0	19.3	21.7	23.6		
	5%	WB	17.5	17.3	16.2	13.6	11.2	9.1	8.6	9.5	11.2	12.8	14.8	16.6		
		MCDB	22.7	22.3	20.7	17.0	14.1	11.4	11.1	12.5	15.4	17.1	19.6	21.4		
	10%	WB	16.4	16.2	15.2	12.6	10.4	8.3	7.7	8.7	10.3	11.6	13.7	15.5		
		MCDB	20.9	20.5	19.1	15.7	12.9	10.4	10.0	11.2	13.9	15.5	17.8	19.7		
(35) Mean Daily Temperature Range	MDBR		10.5	10.5	10.8	10.3	9.3	8.8	9.4	9.7	10.2	10.2	10.0	9.8		
	5% DB	MCDBR	15.5	14.5	14.6	13.7	12.5	11.2	11.6	12.7	13.7	14.2	14.3	13.8		
		MCWBR	7.9	7.8	8.1	8.3	8.2	7.7	7.8	8.5	8.3	8.4	7.7	7.1		
	5% WB	MCDBR	13.6	12.8	13.0	11.9	11.0	10.2	10.4	11.2	12.3	12.9	13.0	12.4		
		MCWBR	7.6	7.2	7.8	7.7	7.7	7.5	7.4	7.9	8.0	8.1	7.5	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.338	0.332	0.312	0.304	0.290	0.288	0.281	0.305	0.319	0.322	0.330	0.333		
	taud		2.525	2.554	2.617	2.612	2.613	2.575	2.604	2.561	2.509	2.522	2.514	2.543		
	Ebn at Noon		981	958	931	860	793	746	795	844	904	958	984	994		
	Edh at Noon		107	98	83	69	56	52	55	71	90	100	107	106		
(44) All-Sky Solar Radiation	RadAvg		5.86	5.17	3.83	2.46	1.48	1.13	1.37	2.10	3.34	4.70	5.73	6.19		
	RadStd		0.45	0.40	0.17	0.16	0.09	0.08	0.09	0.13	0.19	0.27	0.29	0.38		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	+0.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page